

A Novel Ultra-Compact FPGA-Compatible TRNG Architecture Exploiting Latched Ring Oscillators

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Abstract—In this brief we present a novel, ultra-compact, True Random Number Generator (TRNG) architecture and its FPGA implementation. The proposed Latched Ring Oscillator (LRO) TRNG allows the generation of a TRNG bit from a single FPGA Slice. Despite its very compact structure, the proposed LRO-TRNG relies on both meta-stability and accumulated jitter as entropy sources, and exhibits very good results in terms of unpredictability and randomness. The proposed architecture has been implemented on Xilinx Spartan-6 devices and the TRNG performances have been extensively validated under supply voltage and temperature variations. Measurements results have shown that the LRO-TRNG exhibits an estimated entropy of about 7.99834 per bit (according to T8 test of the AIS-31) and a throughput of 0.76 Mbits/s with a 50MHz clock. A comparison against the state of the art shows that the proposed LRO-TRNG outperforms most of the previously published TRNGs, in terms of the ratio between throughput and FPGA resources usage.

Index Terms—TRNGs, field programmable gate array (FPGA), ring oscillators, metastability, jitter.

I. INTRODUCTION

NOWADAYS the society strongly relies on lightweight and portable devices which execute several tasks in a smart and simple way (e.g., smartwatch, smartband smartphone, tablet and laptop). Most of the daily used services, such as data-storage and banking management, relies on authentication protocols, key and token generation [1]. The True Random Number Generators (TRNGs) are key building blocks in security applications and are used in a plenty of key-generation mechanisms, authentication protocols [2], [3], cryptographic algorithms [4], simulation software problems [5] such as Monte-Carlo simulations [6] and also as random masks in side channel attack countermeasures [7]. Due to the huge number of requirements that have to be simultaneously fulfilled, TRNGs are among the most challenging structures to be deployed on portable devices. In fact, most of these architectures demand for a considerable amount of resources which are often not available on consumer devices.

Manuscript received October 5, 2021; accepted October 15, 2021. Date of publication October 20, 2021; date of current version March 15, 2022. This work was supported in part by the European Union (EU) through the ERC Project under Grant 724725 (SWORD). This brief was recommended by Associate Editor A. J. Acosta. (Corresponding author: Riccardo Della Sala.)

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Digital Object Identifier 10.1109/TCSII.2021.3121537

FPGAs have become very popular in prototyping and deployment of embedded-systems and consequently more and more devices exploit their potentials. In order to develop a portable, lightweight and compact TRNG macro that can be easily integrated on such platforms, the building blocks provided within FPGAs have to be considered.

TRNGs usually rely on dynamic entropy sources, such as jitter accumulation, thermal noise [8], and metastability [5], [9], [10]. Nevertheless, exploitable entropy sources on FPGA devices are quite limited. One of the most used entropy sources in TRNG architectures is jitter in ring oscillators (ROs) [4], [11], [12]. The basic principle behind RO-TRNGs is to simultaneously excite ring-oscillators with different nominal frequencies and sampling the XOR of their outputs at a sampling frequency f_s in order to generate a random output sequence which depends on jitter accumulation. The larger is the set of ring-oscillators which are involved, the higher is the entropy of the system, although resources usage and power-consumption are increased too. It has to be noted that, these designs often require post-processing encoding schemes (e.g., Von Neuman encoding [12]) in order to pass the statistical tests, which result in additional FPGA resources and power consumption.

The metastability of NAND-latches has been exploited in [13] as an alternative entropy source in place of ROs jitter. Feedback circuits have been employed in [14] to improve the output sequence's characteristics, thereby impacting on the overall resources consumption. A TRNG with self-control feedback to enhance entropy has been proposed in [5].

An interesting TRNG approach based on Digital Clock Manager (DCM) blocks available in Xilinx FPGAs has been exploited in several works such as, for example, [15], [16]. In [16] the jitter of the FPGA's DCMs and the metastability of memory elements are used as entropy sources. Like most of the FPGA-compatible TRNG implementations, DCM-based ones require additional logic to improve the randomness performance of the extracted sequence.

In this work we propose a novel lightweight TRNG architecture which leverages on the noise characteristics of D-latches and LUTs and exploits both jitter in ROs and D-Latches' metastability as entropy sources. For the implementation we have chosen the Xilinx Spartan-6 XC6SLX16 FPGA platform and we have investigated in-depth all possible interconnections among the elements of the FPGA as well as the characteristics and the physical behaviors of the FPGA built-in blocks. Such investigations allowed us to gain insight into the effect of jitter's accumulation and to fully exploit the metastability, thereby optimizing the TRNG performances in terms of entropy and randomness without sacrificing hardware resources. This brief is structured as follows. In Section II, we introduce the novel LRO-TRNG architecture,

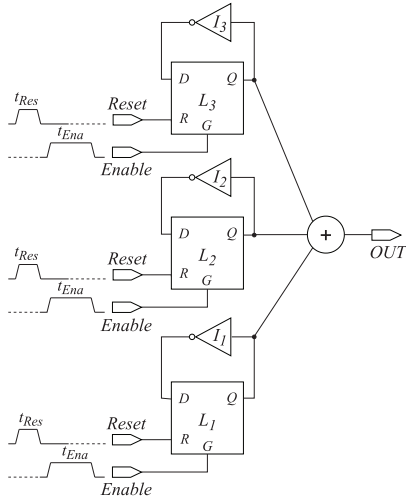


Fig. 1. Proposed TRNG Architecture.

and in Section III we describe its FPGA implementation. In Section IV, we present results on the analysis of the random bitstream, adopting commonly used test suites. In Section V we compare our proposed design with the state-of-the-art of TRNGs. Finally, conclusions are drawn in Section VI.

II. PROPOSED TRNG ARCHITECTURE

The block scheme of the proposed TRNG architecture is shown in Fig. 1. The single TRNG cell exploits three latches L_1 , L_2 and L_3 closed in a ring oscillator configuration through three inverters I_1 , I_2 and I_3 respectively. When a logic '0' is applied to the Gate inputs (G) of the D-Latches, they are in the hold state and are insensitive to variations on the D inputs. On the other hand, when the G inputs of the latches are set to logic '1', the latches become transparent and their outputs Q follow the variations on the D inputs. According to this behavior, when the latches are transparent a free running oscillation comes out, whereas, when they enter in the hold state, the logic value of the output bit is sampled. Since the propagation delay from the D inputs to the Q outputs t_{DQi} , ($i = 1, 2, 3$) of D-Latches L_1 , L_2 and L_3 , and the propagation delay t_{Ivi} , ($i = 1, 2, 3$) of the inverters I_1 , I_2 and I_3 , are dependent on the physical implementation and on the delay of the routing path of the three latches, each RO exhibits an oscillation period (denoted as T_{ROi}) equals to:

$$T_{ROi} = \frac{1}{2 \cdot (t_{DQi} + t_{Ivi})} \quad (1)$$

The operation of the proposed TRNG requires the following excitation sequence:

- 1) Set the *Reset* signals to logic '1' for a time t_{Res} in order to reset the outputs of the three D-Latches to logic '0';
- 2) Set a logic '0' on the *Reset* inputs and set the *Enable* signals to logic '1' for a time t_{Ena} thus enabling the three ROs;
- 3) After a time t_{Ena} , set a logic '0' on the *Enable* input in order to sample the output random bit.

Each bit has to be generated by means of the above *Reset* and *Enable* sequence.

It is evident, from these considerations, that the overall throughput (TP) is limited by the time $t_{Res} + t_{Ena}$ and the

bit-sequence throughput is given by the following equation:

$$TP = \frac{1}{t_{Res} + t_{Ena}} \left[\frac{\text{bit}}{s} \right] \quad (2)$$

As will clarified in the next sections, the entropy of the output random sequence is directly related to the accumulated jitter and therefore to the excitation time t_{Ena} of the ROs. In fact, since an increase of t_{Ena} results in an increase of the entropy and in a reduction of the throughput, the proposed TRNG architecture requires to optimize the trade-off between these two figures of merit. For this purpose, it is important to remark that the behavior of the proposed LRO-TRNG is quite different from the one of conventional RO-TRNGs. In fact, previous works exploit the jitter of the ROs and the metastability of D-flip-flops by sampling the stream started in a single instant. It has to be noticed that in the proposed LRO-TRNG, when the gates are closed (i.e., the G inputs are at logic '0' and the data is sampled) a metastable state can be captured, thus increasing the entropy due to the metastability of D-Latches.

However, since process, supply voltage and temperature (PVT) variations affect the oscillation frequencies of the three LROs of a single cell in the same way (i.e., are seen as common mode variations), the XOR operation between the output of the three D-Latches L_1 , L_2 and L_3 greatly improves the resilience of the proposed TRNG to PVT variations, thus providing very good statistical performances in spite of working condition variations as will be shown in Section IV-B. Another important point to remark is that, if the oscillation frequencies of the three LROs are extremely close to each other, locking phenomena can occur [17], and the statistical performances of the TRNG can be worsened, thus requiring additional post-processing to perform on the output sequence. To avoid these issues, the oscillation frequencies of the three LROs have to be properly unbalanced during the implementation phase by exploiting the different delays available through the different FPGA blocks and routing resources. The whole design flow and routing strategies have been investigated accordingly to [18].

III. LRO-TRNG HARDWARE IMPLEMENTATION

As target FPGA device for the implementation of the LRO TRNG we have chosen the Xilinx Spartan-6 XC6SLX16 [19], using Xilinx ISE 14.7 as a development toolchain. Since the exploitation of D-Latches' metastability and jitter accumulation are directly related to the physical implementation of the routing interconnections and logic-block placement, an in-depth study on the FPGA's hardware architecture has been conducted.

A. LRO-TRNG FPGA Implementation

The architecture of the Xilinx Spartan-6 is composed by a matrix of Configurable Logic Blocks (CLBs), whose intra and inter connections are managed by the Switch-Matrix [20]. Each CLB is composed by two Slices, each of them containing 4 Look-Up-Tables (LUTs) which could be used to implement logic functions. Moreover, each Slice contains 8 Flip-Flops (FFs), four of which can be configured as D-Latches with shared R and G inputs. For further details see the official design guide [19].

The LRO-TRNG has been implemented in a single FPGA Slice by means of four LUTs and three Latches. Several

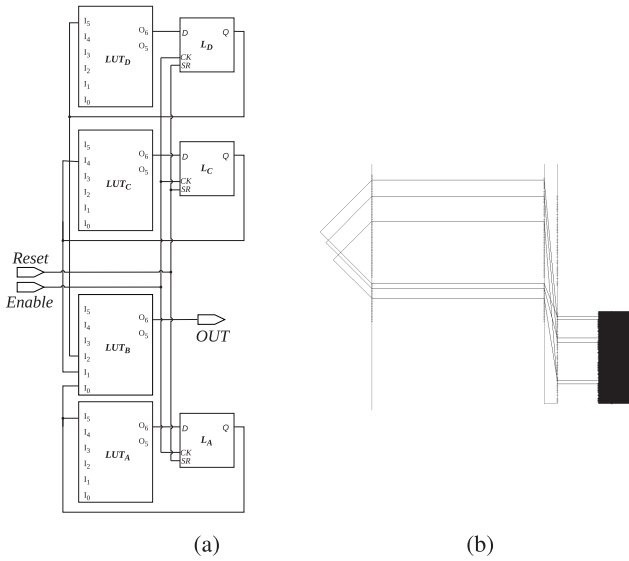


Fig. 2. Slice implementation of the proposed TRNG (a); FPGA-editor's view of the Slice's intra-connections (b).

TABLE I
ESTIMATED RING-OSCILLATOR CHARACTERISTICS

	E.D.* [ps]	E.F. [†] [MHz]
$t_{QI_{5D}}$	306	398
$t_{QI_{5C}}$	697	304
$t_{QI_{5A}}$	427	363

*Estimated Delay through ISE-FPGA-Editor;

[†]Estimated Frequency through ISE-FPGA-Editor and Data-Sheet [22].

routing configurations have been investigated and the one that gives the best performances in terms of randomness and entropy has been selected and depicted in Fig. 2(a). As can be observed in Fig. 2(a) the LUTA, LUTC and LUTD have been employed to implement the inverters depicted in Fig. 1, whereas the three flip-flops configured as latches are FF_A , FF_C and FF_D . Finally the three outputs of the D-Latches are XOR-ed by means of LUTB. The FPGA-Editor floorplan view of the feedback-connections of the LRO-TRNG macro-cell is reported in Fig. 2(b). An in-depth study of the switch-matrix block and the Placement Constraints provided by the Xilinx Guide [21] allowed us to choose the intra-connections resulting in optimal randomness performance. Delays of the routed paths of LUTA, LUTC and LUTD are denoted as $t_{QI_{5A}}$, $t_{QI_{5C}}$ and $t_{QI_{5D}}$ respectively are reported in Table I. The nominal value of the delay between the LUTs' inputs and the D-Latches' outputs is about 950ps,¹ according to [22]. Table I reports also the estimated oscillation frequencies of the ROs in each Slice. As can be observed the three frequencies are in the range of [300,400] MHz; in this estimation we have also taken into account the fan-in effect of the LUTB. Another important feature of the proposed implementation is the possibility for the three D-Latches to share the same *Reset* and *Enable* signals, thus improving the synchronism of the excitation sequence.

B. Test-Bench

The block scheme of the testbed to characterize the TRNG-cell is reported in Fig. 3. A 1024 bits First-In First-Out

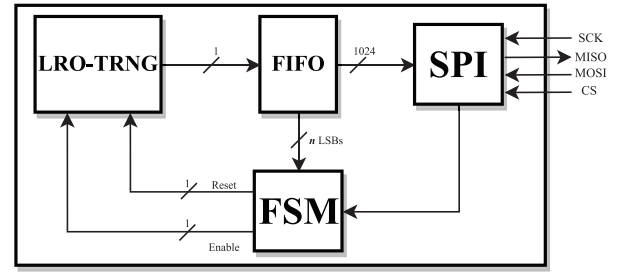


Fig. 3. Block scheme of the TRNG validation testbed.

(FIFO) block is used to collect the TRNG bits. The core voltage of the FPGA (XC16SXL16) has been supplied through the Teledyne T3PS43203P programmable power supply unit through a dedicated connection on the board in order to allow the investigation of the TRNG performance under supply voltage variations. A system clock of 50MHz has been used in all the experiments. The FPGA board is controlled through a daughter board mounting a FT232H chip, which serves as USB-SPI interface, stimulated by custom *Python* scripts. We refer hereon to the duration of the *evaluation phase* t_{Res} and t_{Ena} in number of clock cycles, considering the system clock period as timing reference. Since 20ns are enough to completely reset the LROs, the duration of t_{Res} has been set to 1 Clock-Cycle. The duration of t_{Ena} can be varied by means of the SPI interface and a Finite State Machine (FSM). A feedback strategy to randomly vary the less significative bits (LSBs) setting the duration of t_{Ena} has been implemented in order to improve the randomness of the TRNG as will be better described in Section IV-A. By using this simple but effective setup, we have been able to characterize the proposed TRNG under different stimuli conditions, revealing some interesting features when different t_{Ena} values are considered.

IV. MEASUREMENT RESULTS

In this Section we report the extensive characterization that we have carried out on the LRO-TRNG by measuring the Entropy in different excitation conditions and by executing the commonly used test suites for TRNGs validation: the NIST SP 800-22 suite [23] and the AIS-31 [24].

A. LRO-TRNG Entropy Characterization and Performance Evaluation

The TRNG cell has been studied by means of the setup shown in Fig. 3. The duration of t_{Ena} has been varied by means of the SPI interface and the FSM in order to fully characterize the extracted randomness. The analysis of the sequences has been carried out without any additional post-processing to evaluate the intrinsic quality of the random bit-stream. As a first step, the Shannon entropy of the architecture has been measured for different deterministic values of t_{Ena} in order to identify the minimum value required by the LRO-TRNG cell to generate sufficient entropy. More specifically, the Shannon entropy has been evaluated by considering sequences of 102,400 bits and for t_{Ena} values ranging from 1 to 128 clock cycles, and the results are shown in Fig. 4(a). We have found that the Shannon Entropy starts to approach the ideal value for t_{Ena} longer than about 30 clock cycles, confirming the jitter accumulation effect and highlighting the need

¹The speedgrade -2 has been considered in the delay estimation.

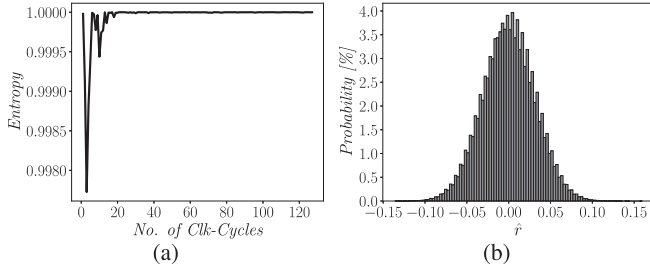


Fig. 4. Entropy of the sequence vs t_{Ena} (expressed as number of clock cycles) (a); Pearson's correlation coefficient between the sequences extracted from the FIFO (b).

to address the trade-off between Entropy and throughput. For t_{Ena} equal to 30 clock cycles the Shannon Entropy has been measured to be 0.99999474 per bit. As a second set of experiments, we have exploited the feedback strategy allowed by the testbed in Fig. 3, in order to extract the four LSBs of the digital word which set the duration of t_{Ena} from the 1024 bits FIFO containing the random stream. Results obtained in this second set of measurements have shown that a very good trade-off between entropy and throughput can be achieved for t_{Ena} equal to $48 + x$ clock cycles, where x is the random value (in the range of $[0, 15]$) represented by the 4 LSBs in the feedback path. The measured Shannon Entropy in this condition has been found to be 0.99999975 per bit.

In order to evaluate the *TP* of the proposed TRNG we start from the aforementioned stimulation strategy, which assumes t_{Res} equal to 1 clock cycle and the worst case t_{Ena} equal to 63 clock cycles. From these values and considering two additional clock cycles between two consecutive excitation phases, the total throughput of the proposed TRNG implementation has been found to be about 0.76 Mbits/s for an operating frequency (*OF*) equal to 50MHz.

B. Randomness Evaluation

As an additional test, we have measured the correlation of 1000 sequences of 1024 bits extracted by collecting multiple start-up and reset phases and by computing the Pearson's correlation coefficient (as reported in Eq. (4) in the Appendix) for all the permutations. The distribution of the correlation coefficient values is depicted in Fig. 4(b). We have found that the mean value of the correlation is about $\mu \approx 5.19 \cdot 10^{-5}$ whereas its standard deviation is $\sigma \approx 0.03$, thus confirming that the extracted bits can be considered as uncorrelated. Finally, to assess the randomness's quality of the sequences extracted from the LRO-TRNG cell, NIST SP800-22a tests [23] have been performed on 100 bitstreams of 1 Mbit each. Results are reported in Table II. As can be observed, the sequences pass all the tests with a pass-rate greater than the minimum required (i.e., 96/100 assuming the significance level $\alpha = 0.01$). This is a remarkable result since the sequences at the output of the LRO-TRNG have been tested without adopting any post-processing technique. In addition, we have performed the analysis also considering temperature and power supply voltage variations through the setup reported in [18]. Two additional datasets have been collected at $\pm 10\%$ of the nominal supply voltage (1.2V) and the NIST's test have been reported in Table III with and without considering feedback strategies. It is evident that the t_{ena} modulation through the 4 LSBs significantly improves the quality of the extracted bitstreams. At 1.08V, the proposed

TABLE II
NIST SP800-22A TEST RESULTS AT 1.2V

Name of the Test	Without feedback			With feedback		
	p-value	pass-rate	result	p-value	pass-rate	result
Frequency Test	0.052	97/100	✓	0.076	100/100	✓
Block Frequency Test	0.043	98/100	✓	0.086	99/100	✓
Cumulative Sums [†]	[0.014,0.022]	99/100	✓	[0.335,0.637]	100/100	✓
Runs Test	0.074	100/100	✓	0.182	100/100	✓
Longest Run	0.341	100/100	✓	0.851	99/100	✓
Rank	0.421	97/100	✓	0.817	98/100	✓
FFT	0.019	96/100	✓	0.097	98/100	✓
Non Overlapping	[0.017,0.298]	[96,100]/100	✓	[0.014,0.999]	[96,100]/100	✓
Overlapping	0.421	99/100	✓	0.798	98/100	✓
Universal	0.012	97/100	✓	0.726	99/100	✓
Approximate Entropy	0.045	100/100	✓	0.335	100/100	✓
Random Excursions [†]	[0.017,0.797]	[55/57]/57 [‡]	✓	[0.060,0.672]	[65/68]/68 [‡]	✓
Random Excursions Variant [†]	[0.024,0.994]	[55/57]/57 [‡]	✓	[0.030,0.976]	[65/68]/68 [‡]	✓
Serial	0.374	98/100	✓	0.699	100/100	✓
Linear Complexity	0.019	96/100	✓	0.334	96/100	✓

[†] The minimum pass-rate is 64/68; [‡] The minimum pass-rate is 54/57;

[†] When multiple value are given it has been reported the values' range attained.

TABLE III
NIST SP800-22A TEST RESULTS VS V_{DD} VARIATIONS

Name of the Test	1.08 V			1.32 V		
	p-value	pass-rate	result	p-value	pass-rate	result
Frequency Test	0.658	99/100	✓	0.029	96/100	✓
Block Frequency Test	0.679	99/100	✓	0.494	100/100	✓
Cumulative Sums [†]	[0.067,0.760]	99/100	✓	[0.001,0.003]	[96,98]/100	✗
Runs Test	0.554	99/100	✓	0.798	99/100	✓
Longest Run	0.956	99/100	✓	0.637	99/100	✓
Rank	0.456	98/100	✓	0.596	99/100	✓
FFT	0.554	100/100	✓	0.616	99/100	✓
Non Overlapping	[0.014,0.999]	[97,100]/100	✓	[0.018,0.991]	[97,100]/100	✓
Overlapping	0.202	98/100	✓	0.834	99/100	✓
Universal	0.350	100/100	✓	0.059	100/100	✓
Approximate Entropy	0.384	97/100	✓	0.456	99/100	✓
Random Excursions [†]	[0.204,0.994]	[60,61]/61 [‡]	✓	[0.103,0.760]	[53,55]/55 [‡]	✓
Random Excursions Variant [†]	[0.070,0.971]	[60,61]/61 [‡]	✓	[0.091,0.964]	[54,55]/55 [‡]	✓
Serial [†]	[0.319,0.384]	[97,100]/100	✓	[0.046,0.817]	[97,99]/100	✓
Linear Complexity	0.596	100/100	✓	0.130	98/100	✓

[†] The minimum pass-rate is 58/61; [‡] The minimum pass-rate is 52/55;

[†] When multiple value are given it has been reported the range of the attained values;

TABLE IV
AIS-31 ESTIMATED BYTE-ENTROPY (TEST T8)

Environmental Condition	1.08V , 25°C	1.2V, 25°C	1.32V, 25°C	1.2V, 50°C	1.2V, 75°C
Estimate Byte-Entropy	7.99652	7.99834	7.99387	7.99787	7.99819

architecture passes all tests, while at 1.32V, the LRO-TRNG exhibits a slight degradation of the quality of the raw random bitstream, as only one test does not formally reach the required minimum values (i.e., the Cumulative Sums test doesn't show a p-value greater or equal than 0.01). As further evaluation, we have also performed the AIS-31 tests on the collected raw samples. In this case, all AIS-31 tests have been passed for all the three considered power supply voltage values. Estimated byte-entropy values in test T8 procedure B are reported in Table IV. It is worth noting that also for AIS-31, the T8 tests shows a slight degradation of the quality of the generated random bitstream at 1.32V. Furthermore, it has been investigated the bitstream quality under temperature variations by means of a climatic chamber. It was found that in any environment condition all AIS-31 tests pass and no significant entropy degradation occurs as highlighted from T8 tests reported in Table III.

V. COMPARISON WITH THE STATE-OF-THE-ART

The proposed LRO-TRNG has been compared against other FPGA-Compatible TRNGs recently published in the technical literature and the results of this comparison are reported in

TABLE V
COMPARISON TABLE

Mode Ref	Area [◊]			TP [Mbit/s]	O.F. [MHz]	FOM [$\frac{\text{bits}}{\text{Slice}}$]	Byte Entropy [†]	Platform
	LUTs	Flip-Flops	Slices					
LRO This Work	4	3	1	0.76	50	$15.15 \cdot 10^{-3}$	7.9983	Spartan-6
RO [12]	528	177	270	6	24	$0.26 \cdot 10^{-3}$	7.9946	Spartan-3A
RO [25]	712	753	-	3.2	-	-	-	Spartan-3E
RO [4]	10	5	3 [‡]	1.15	100	$3.83 \cdot 10^{-3}$ [‡]	7.9760 [‡]	Spartan-6
RO [7]	521	131	-	2.57	-	-	7.9920	Spartan-6
Meta-stability [10]	-	-	580	12.5	100	$0.22 \cdot 10^{-3}$	-	Virtex-4
Meta-stability [9]	128	-	-	2	-	-	-	Virtex-5

* Flip-Flops used as latches; [◊] Area of the TRNG cell without considering control logic; [†] Estimated considering the Xilinx Spartan-6 CLBs [20];

[‡] Estimated; [‡] Estimated with T8 test of the AIS-31.

Table V. According to Table V, the proposed implementation attains the best results in terms of resources usage (Slices) and estimated entropy per byte with T8 test of the AIS-31. In fact, only the TRNG reported in [4] is comparable to the proposed one in terms of resources consumption. However, computing the ratio between *TP* and *OF*, it can be seen that the proposed LRO-TRNG exhibits an higher value of this ratio than the TRNG in [4]. To better compare different designs with different operating frequency and implemented on different FPGA devices, we introduce the following Figure Of Merit (*FOM*) which summarizes the performance in terms of *TP* over Slices and *OF*:

$$FOM = \frac{TP}{OF \cdot Slices} \left[\frac{\text{bits}}{\text{Slice}} \right] \quad (3)$$

From Table V we can observe that the proposed LRO-TRNG outperforms all the other TRNG in terms of the *FOM* reported in Eq. (3).

VI. CONCLUSION

A novel TRNG architecture that leverages the benefits of both the jitter of ring oscillators and the D-Latches' metastability has been proposed. A feedback strategy to randomly set the 4 LSBs of the control word defining the excitation time has been exploited to enhance the randomness and increase the entropy. The randomness of the raw response without post-processing techniques has been tested with NIST's and AIS-31 test suites, showing that also when supply voltage variations are considered the TRNG is able to fulfill the test requirements. The proposed LRO-TRNG allows to generate 1 bit per Slice with a throughput of about 0.76 Mbits/s. Comparison against the state-of-the-art has shown that the proposed implementation exhibits the best performance in terms of throughput per Slice with respect to all the previously published FPGA-compatible TRNGs resulting the most compact TRNG ever integrated on Xilinx FPGAs.

APPENDIX

Given two sequences of bits x, y with length N , the Pearson Correlation Coefficient could be defined as:

$$\hat{r} = \frac{\sum_{i=0}^{N-1} (x_i - \bar{x}) \cdot (y_i - \bar{y})}{\sqrt{\sum_{i=0}^{N-1} (x_i - \bar{x})^2 \cdot \sum_{i=0}^{N-1} (y_i - \bar{y})^2}} \quad (4)$$

referring to the average value of x and y with $\bar{x}$ and $\bar{y}$ respectively and with $\hat{r} \in [-1, 1]$.

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